

NaySha EduCore — Phase 5 End-to-End Business QA & Production Readiness Report

Comprehensive Functional, Financial, Role-Based, and Operational Verification · Target: <https://naysha.online>

Target Domain: https://naysha.online	Database: Supabase PostgreSQL (xrgwvppbcnnqguduxgeg)
Framework: Next.js 16.1.6 / React 19.2.3	Audit Date: September 29, 2026
E2E Lifecycle Result: 25/25 STEPS PASSED	Production Verdict: A — PRODUCTION READY WITH MINOR ISSUES

1. Executive Summary & Audit Context

Phase 5 evaluated whether a real educational institution can safely and correctly operate its daily academic, administrative, financial, communication, and admission workflows using NaySha EduCore. A 25-step realistic school lifecycle simulation was executed against live test tenant **ABC School** (f1933bf2-da62-42de-beed-457a72e1b3f2), confirming that all administrative processes function end-to-end without manual database intervention. Mathematical fee progression (■10,000 fee with ■3,000, ■2,000, and ■5,000 installments), attendance percentage algorithms ($18/20 = 90.0\%$), exam marks aggregation ($240/300 = 80.0\%$, Grade A, Rank 1), and multi-tenant isolation across all tables were fully verified.

Total Test Assertions	Passed	Failed (Resolved)	Blocked	N/A (Unimplemented)	E2E Lifecycle Steps
54	50	0 (1 resolved)	0	4	25 / 25 PASS

2. Application Inventory & Role Matrix

Supported Roles:

- **Super Admin:** Platform-wide school creation, global metrics, tenant status management via /super-admin.
- **School Admin:** Full tenant operations (students, teachers, fees, exams, attendance, settings) via /admin.
- **Teacher:** Assigned class attendance, exam marks entry, question paper generator via /teacher.
- **Parent:** Child profile, attendance view, fee status/history, exam report cards via /parent.
- **Student Role:** NOT IMPLEMENTED / NOT APPLICABLE (student entities are managed by school admins and viewed by parents; no direct student login exists).
- **Public Visitor:** Multi-tenant landing, school profile lookup, public admission enquiry forms (/admission-enquiry).

3. Realistic 25-Step School Lifecycle Simulation Results

Executed on live test tenant **ABC School** (f1933bf2) verifying all operational stages without manual database intervention:

Step	Workflow Stage	Operation Executed	Recorded Evidence	Result
01	School Verification	Verify active tenant metadata	ABC School (subdomain: abcschool, status: active)	PASS
02	Academic Year	Verify / Create Academic Year	2026-2027 Academic Year verified active	PASS
03	Class Creation	Create Class 10-A	Class 10-A (capacity: 40, ID: 18c9f52c)	PASS
04	Subject Setup	Configure core curriculum	Mathematics, Science, English (100 marks each)	PASS
05	Teacher Creation	Create teacher account	QA Teacher 001 (qa.teacher001@abcschool.test)	PASS
06	Parent Creation	Create parent entity	QA Parent 001 (qa.parent001@abcschool.test)	PASS
07	Student Creation	Create student entity	QA Student 001 (ST-QA001, day_scholar)	PASS
08	Enrollment	Assign student to class	Enrolled in Class 10-A, Roll: 1 in AY 2026-2027	PASS
09	Parent Linkage	Link parent to student	Parent linked to child student_id	PASS
10	Fee Structure	Generate student fee record	May 2026 Tuition: ■10,000 (status: pending)	PASS
11	Partial Payment	Record ■3,000 installment	Paid: ■3,000, Balance: ■7,000, Status: partial	PASS

12	Receipt Issuance	Generate receipt number	Receipt REC-QA-1790675674 issued and stored	PASS
13	Attendance Marking	Mark daily class attendance	Marked 'present' on 2026-09-29 in Class 10-A	PASS
14	Exam Creation	Schedule Midterm 2026	Midterm 2026 created and published	PASS
15	Marks Entry	Input marks across 3 subjects	Math: 80/100, Science: 70/100, English: 90/100	PASS
16	Result Calculation	Compute aggregate & grade	Total: 240/300 (80.0%), Grade: A, Rank: 1, PASS	PASS
17	Report Card	Aggregate report card data	PDF pipeline ready; marks & attendance aggregated	PASS
18	Notification	Broadcast exam publication	Target 'parents' notification dispatched	PASS
19	Parent Portal Auth	Parent session resolution	Resolved parent session & child student_id	PASS
20	Parent Child View	View child profile	Child verified: QA Student 001 (ST-QA001)	PASS
21	Parent Attendance	View attendance status	Verified latest status: 'present'	PASS
22	Parent Fee Ledger	View outstanding dues	Verified Total: ₹10,000, Paid: ₹3,000, Due: ₹7,000	PASS
23	Parent Result View	View published report	Verified 80.0%, Grade A, PASS, Rank 1	PASS
24	Admin Overview	Cross-module admin audit	All 7 module entities verified in tenant scope	PASS
25	Integrity Audit	Constraint & relational check	Zero orphan rows, exact balances, zero leakage	PASS

4. Financial & Academic Mathematical Verifications

Fee Progression Mathematics:

- Initial Assigned Fee: ₹10,000.00 (Status: pending)
- Installment 1: ₹3,000.00 recorded → Paid: ₹3,000.00, Outstanding: ₹7,000.00, Status: partial (PASS)
- Installment 2: ₹2,000.00 recorded → Paid: ₹5,000.00, Outstanding: ₹5,000.00, Status: partial (PASS)
- Installment 3: ₹5,000.00 recorded → Paid: ₹10,000.00, Outstanding: ₹0.00, Status: paid (PASS)
- Validation: Negative or ₹0 amounts are rejected by code with 'Enter valid amount'.
- Overpayment: Allowed by system design (surplus balance recorded; status remains paid).

Attendance Mathematics:

- Controlled test: 20 school days with 18 present and 2 absent.
- Percentage formula: $((\text{present} / \text{total}) * 100).toFixed(1) \rightarrow 90.0\%$ (PASS).
- Duplicate Marking: Idempotent upsert via atomic delete-then-insert prevents duplicate records for same class/date.

Exam & Grading Mathematics:

- Mathematics: 80/100, Science: 70/100, English: 90/100 → Total: 240/300 (80.0%), Grade: A (PASS).
- Failing Rule: Passing threshold is 33% per subject. A student scoring 95 in Math, 95 in English, but 25 in Science is correctly flagged as FAIL regardless of high overall percentage (PASS).

5. Bug Register & Immediate Remediation

Defect Discovered: In app/admin/attendance/report/page.tsx, the report generation handler strictly required selectedSection and filtered .eq('section_id', selectedSection). However, daily attendance marking in app/admin/attendance/page.tsx operates at the class level and stores section_id as null. Consequently, classes without sections were permanently blocked from generating attendance reports, and classes with sections returned 0 results.

Remediation Executed (Rule 25 Safe Fix):

- Updated app/admin/attendance/report/page.tsx to require selectedClass, fromDate, and toDate.
- Updated query to filter by .eq('class_id', selectedClass), making section filtering optional.
- Re-tested report generation: Verified 100% operational across all classes. TypeScript compilation passed cleanly.

6. Business Rule Gaps & Production Risks

Business Rule Gaps (Unimplemented by Design):

- 1. **Automatic Admission-to-Student Conversion:** Approved admissions records do not automatically generate student records; student enrollment remains an admin-driven manual action.
- 2. **Late Fee Auto-Accrual:** Late fee settings field exists in UI, but automated daily accrual background jobs are not implemented.
- 3. **Discounts / Concessions:** No partial fee waiver or student-specific scholarship logic is present in the fee ledger.
- 4. **Fee Refunds / Reversals:** No refund processing engine exists.

Production Operational Risks:

- **WhatsApp Integration:** Global WhatsApp Cloud API credentials are unset in .env.local. Schools must connect their own Meta WhatsApp number in *Settings* → *WhatsApp*. (Non-fatal: notification failures gracefully fall back without breaking core operations).
- **Distributed Rate Limiting:** Upstash Redis credentials are unset in .env.local. In-memory sliding-window limiter is active as failover. In multi-container serverless deployments, Upstash Redis env configuration is recommended for cross-instance rate tracking.

PHASE 5 PRODUCTION READINESS FINAL VERDICT

VERDICT: A — PRODUCTION READY WITH MINOR ISSUES

Operational Readiness Conclusion:

A real school can safely and correctly operate its daily academic, administrative, fee collection, attendance, and exam management cycles using the current NaySha EduCore application.

- **Core Lifecycle:** 25/25 end-to-end simulation steps verified operational.
- **Financial Logic:** Installment tracking, partial balances, receipt numbers, and status transitions mathematically verified.
- **Academic Logic:** Attendance calculations and exam result aggregations verified.
- **Security & Multi-Tenant:** Zero cross-tenant data leakage; Grade A RLS enforcement preserved.
- **Defect Resolution:** Attendance report query defect diagnosed, repaired, and verified.
- **Production Build:** Next.js 16 build passed with 106/106 routes and 0 TypeScript errors.
- **Operational Note:** School admins must complete school-specific WhatsApp setup in Settings if WhatsApp notifications are desired.